DS	**Session**	Arrays	**Question Information**	Level 1 Challenge 22

Problem Description: Ram has provide inputs two numbers 'p' and 'q' to Sakthi. He wants to creates a matrix of size p x q (p rows and q columns) in which every elements is either Y or 0. The Ys and 0s must be filled alternatively, the matrix should have outermost rectangle of Ys, then a rectangle of 0s, then a rectangle of Ys, and so on..

Constraints:

1 <= p, q <= 1000

Input Format:

Input lines must be how many rows and columns in that matrix, also values must be separate space.

Output Format:

Print the output in a separate lines.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
6 7
```

Test Case 2

INPUT (STDIN)

```
5 8
```

EXPECTED OUTPUT

```
Y Y Y Y Y Y Y
Y 0 0 0 0 0 Y
Y 0 Y Y Y 0 Y
Y 0 Y Y Y 0 Y
Y 0 0 0 0 0 Y
Y Y Y Y Y Y Y
```

EXPECTED OUTPUT

```
Y Y Y Y Y Y Y Y
Y 0 0 0 0 0 0 Y
Y 0 Y Y Y Y 0 Y
Y 0 0 0 0 0 0 Y
Y Y Y Y Y Y Y Y
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
while(top<=bottom &&
right>=left)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

12

Test Case 2

TOKEN COUNT

290

Test Case 3

NLOC

52

Code

solution.c

```c
#include <stdio.h>

int main() {
    int p, q;
    if (scanf("%d %d", &p, &q) != 2) return 0;
    char mat[100][100];
    int top = 0, bottom = p - 1, left = 0, right = q - 1;
    char ch = 'Y';
    while (top <= bottom && right >= left) {
        for (int i = left; i <= right; i++) mat[top][i] = ch;
        for (int i = top; i <= bottom; i++) mat[i][right] = ch;
        for (int i = right; i >= left; i--) mat[bottom][i] = ch;
        for (int i = bottom; i >= top; i--) mat[i][left] = ch;
        top++; bottom--; left++; right--;
        ch = (ch == 'Y') ? 'O' : 'Y';
    }
    for (int i = 0; i < p; i++) {
        for (int j = 0; j < q; j++) {
            printf("%c ", mat[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

Output

Problems Output Debug Console Terminal Ports powershell

```
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
6 7
Y Y Y Y Y Y Y
Y O O O O O Y
Y O Y Y Y O Y
Y O Y Y Y O Y
Y O O O O O Y
Y Y Y Y Y Y Y
PS C:\C Programming>
```